SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2618–2638; original printed p. 88; physical source-PDF p. 78; recomposed running p. 70. The French and printed line-2632 display has an internally inconsistent sign; this bounded target applies and discloses a mathematically forced editorial emendation. The French authority itself remains unchanged, and the compiled PDF is not an independent original witness. Line 2639 is blank; Section 2 begins at line 2640 and is excluded.

The initial term of the first spectral sequence of the bicomplex $\underline{\mathrm{F}}Q^{\bullet\bullet}$ is

$$E_2^{p,q} = {}'H^p({}''H^q(\underline{\mathrm{F}}Q^{\bullet\bullet})).$$

For every $p \in \mathbf{Z}$, CA^p is injective. By the hypothesis on $\underline{\mathrm{F}}$, the functor (restricted to the category of finitely generated modules)

$$N \longmapsto \underline{\mathrm{F}}\,\mathrm{Hom}(N, CA^p)$$

is exact. Hence one obtains isomorphisms

$$''H^q(\underline{\mathrm{F}}\,\mathrm{Hom}(L'^{\bullet}, CA^p)) \cong \underline{\mathrm{F}}\,\mathrm{Hom}(H^{-q}(L'^{\bullet}), CA^p).$$

By the definition of $\mathrm{Ext}^*_{\underline{\mathrm{F}}}$ as the right-derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$, one therefore obtains isomorphisms

$$E_2^{p,q} \cong \mathrm{Ext}^p_{\underline{\mathrm{F}}}(H^{-q}(L'^{\bullet}), A).$$

Source note. The French TeX and printed display have $H^q(L'^{\bullet})$ in the last formula. The preceding displayed isomorphism, the statement of Lemma 1.2, and the identity below require $H^{-q}(L'^{\bullet})$. This target therefore uses H^{-q} as a transparent editorial emendation. It is not a restoration from an independent original scan; the French authority remains unchanged pending editorial action.

Now

$$L'^{\bullet} = \mathrm{Hom}^{\bullet}(L^{\bullet}, A^{\bullet}),$$

where $L^{\bullet}$ is a projective resolution of M; hence there are isomorphisms

$$\mathrm{Ext}^q(M, A) \cong H^q(L'^{\bullet}).$$

Substituting $-q$ in this last identity gives the conclusion. $\square$